

VULNERABILITY ANALYSIS - FOOTPRINTING AND RECONNAISSANCE

Ocean.jpg

ExifTool Output

Structured ExifTool output grouped for easier reading on phones and smaller screens.

Text50 tags

ExifTool1 tags

ExifToolVersion13.42

System7 tags

FileNameocean.jpg

Directory/

FileSize42860

FileModifyDate2026:04:03 12:01:08+00:00

FileAccessDate2026:04:03 12:01:08+00:00

FileInodeChangeDate2026:04:03 12:01:08+00:00

FilePermissions100000

File10 tags

FileTypeJPEG

Raw ExifTool Output

[ExifTool] ExifToolVersion : 13.42

[System] FileName : ocean.jpg

[System] Directory : /

[System] FileSize : 42860

[System] FileModifyDate : 2026:04:03 12:01:08+00:00

[System] FileAccessDate : 2026:04:03 12:01:08+00:00

[System] FileInodeChangeDate : 2026:04:03 12:01:08+00:00

[System] FilePermissions : 100000

[File] FileType : JPEG

[File] FileTypeExtension : JPG

[File] MIMEType : image/jpeg

[File] Comment : THIS IS THE HIDDEN FLAG

[File] ImageWidth : 640

[File] ImageHeight : 425

[File] EncodingProcess : 2

[File] BitsPerSample : 8

[File] ColorComponents : 3

[File] YCbCrSubSampling : 2 2

[JFIF] JFIFVersion : 1 1

[JFIF] ResolutionUnit : 1

[JFIF] XResolution : 72

[JFIF] YResolution : 72

[ICC-header] ProfileCMType : icms

[ICC-header] ProfileVersion : 528

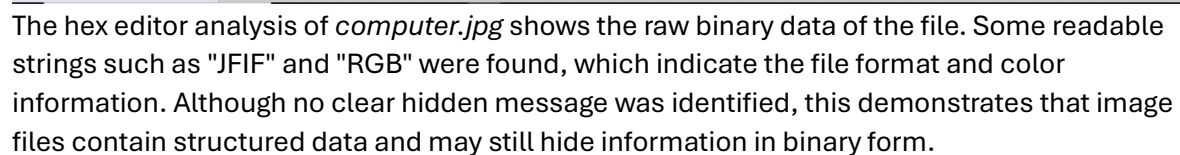
[ICC-header] ProfileClass : mntr

[ICC-header] ColorSpaceData : RGB

[ICC-header] ProfileConnectionSpace : XYZ

[ICC-header] ProfileDataTime : 2012:01:25 03:41:53

Computer.jpg



Dog.jpg

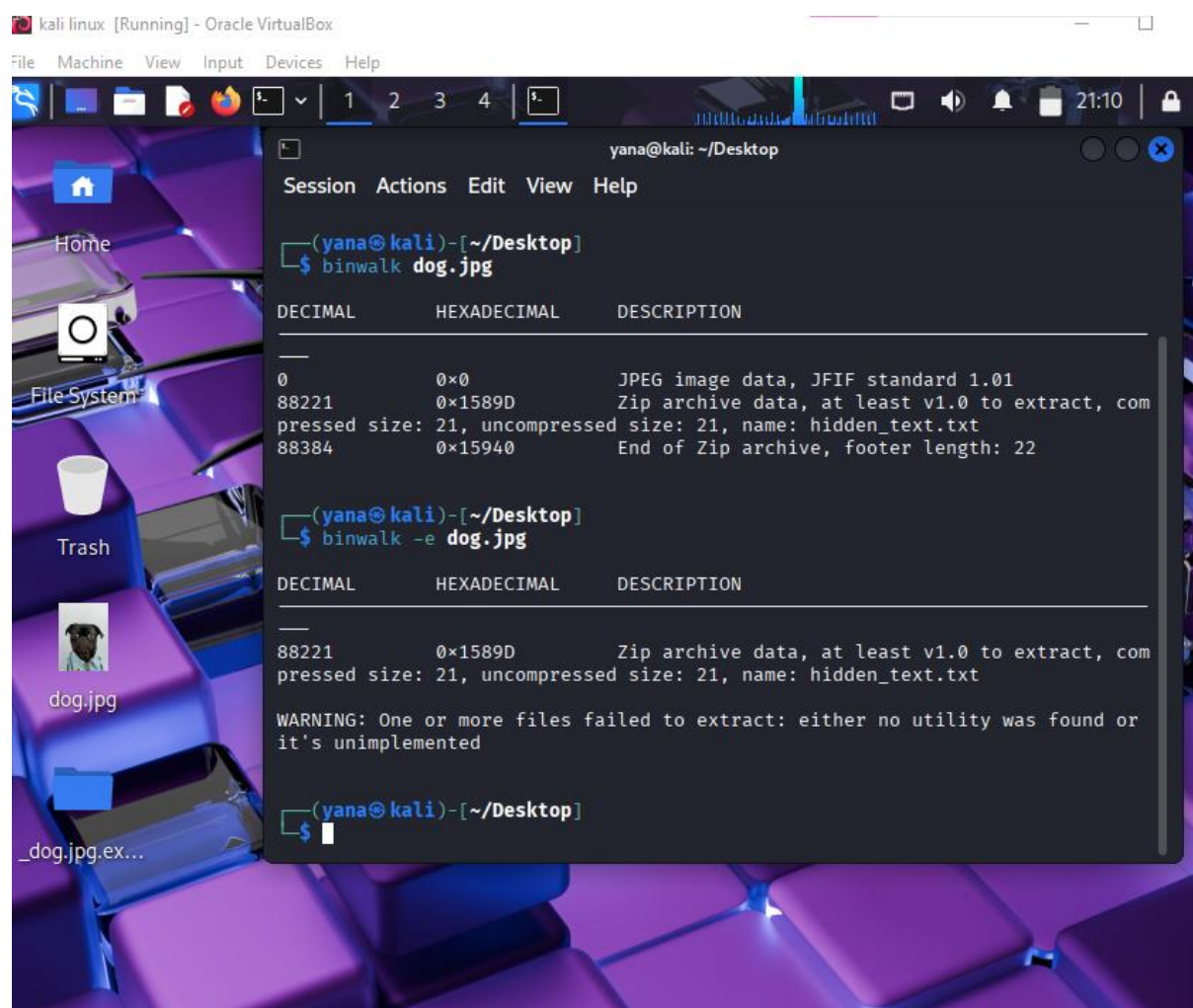
```
(yana@kali)-[~]
$ cd Desktop

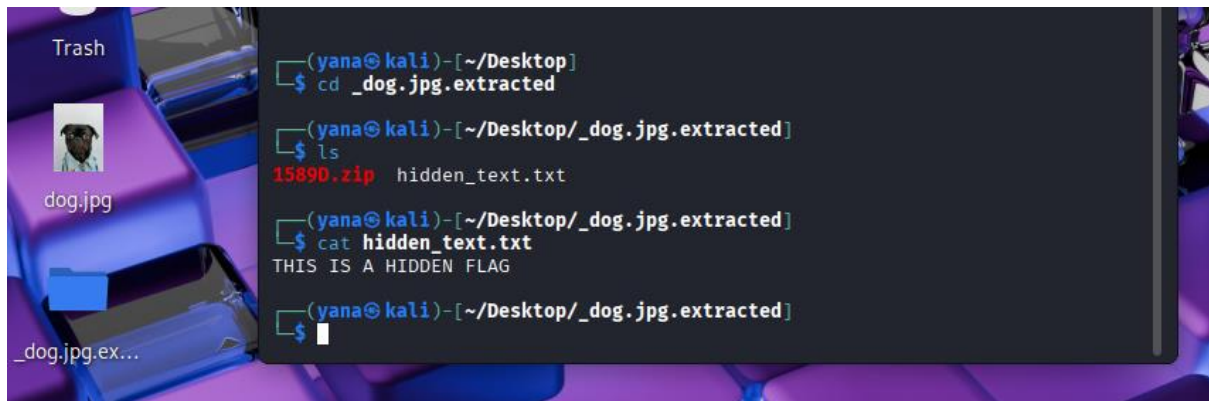
(yana@kali)-[~/Desktop]
$ ls
dog.jpg

(yana@kali)-[~/Desktop]
$ binwalk dog.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, JFIF standard 1.01
88221	0x1589D	Zip archive data, at least v1.0 to extract, compressed size: 21, uncompressed size: 21, name: hidden_text.txt
88384	0x15940	End of Zip archive, footer length: 22

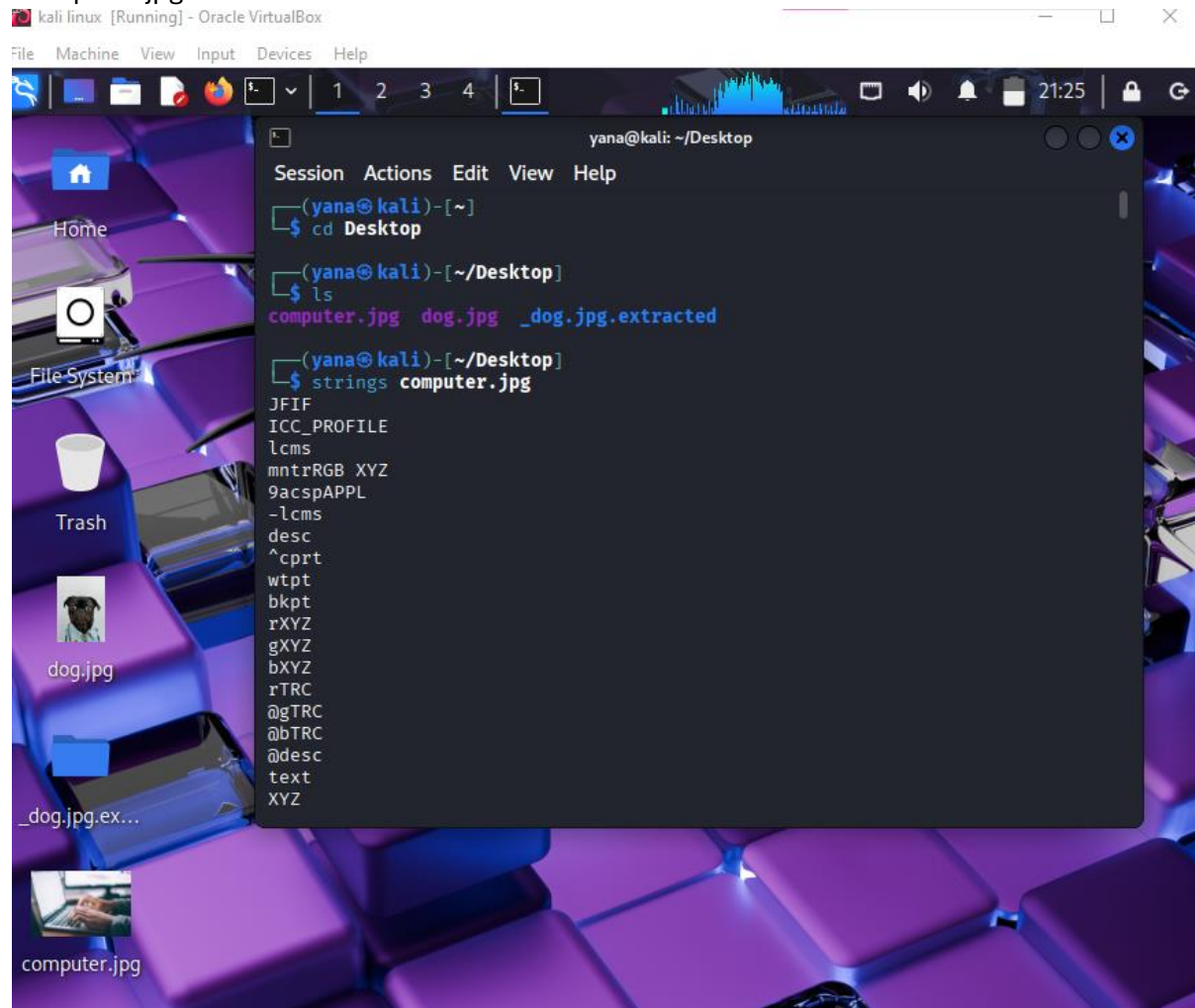
```
(yana@kali)-[~/Desktop]
$
```





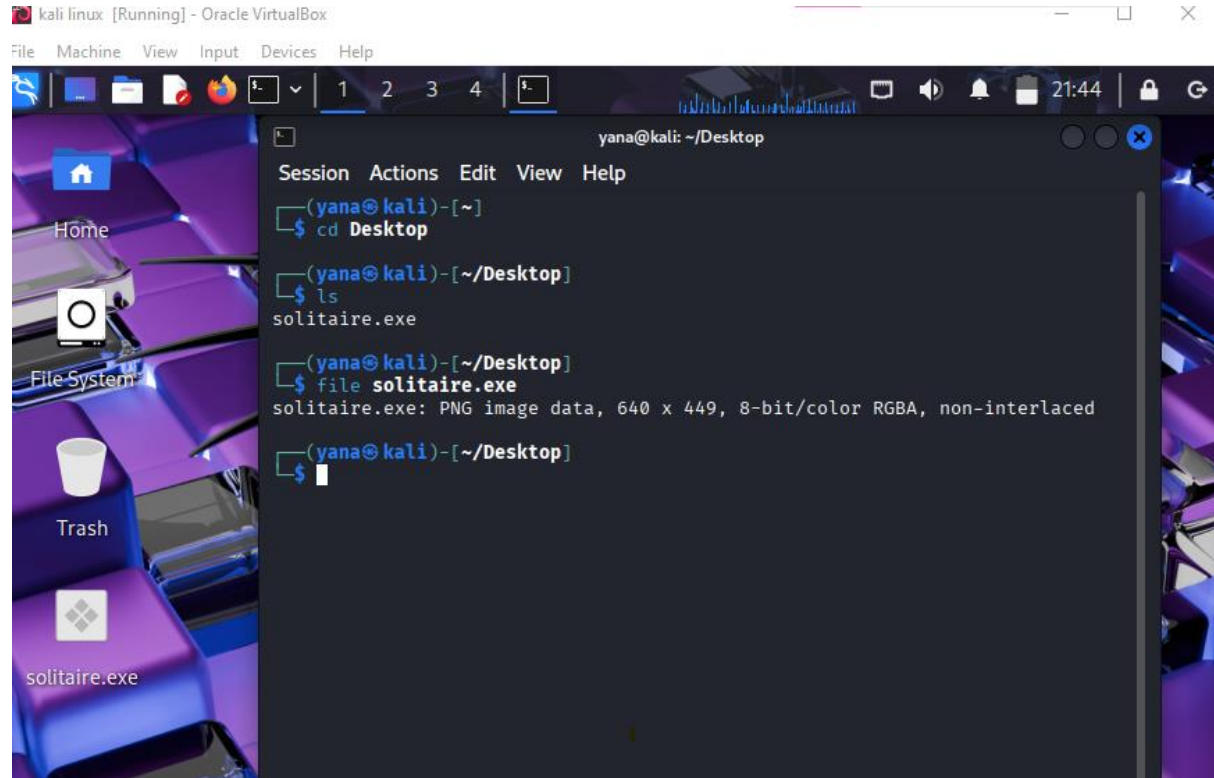
The binwalk analysis of *dog.jpg* detected an embedded zip archive within the image. After extraction, a hidden file named "hidden_text.txt" was found. Viewing the contents of this file revealed hidden information, demonstrating that images can be used to conceal data using steganography techniques.

Computer.jpg



The strings command was used to extract readable text from *computer.jpg*. Some readable strings such as "JFIF", "ICC_PROFILE", and "mntrRGB XYZ" were found, indicating that the image file contains embedded data and metadata. Although most of the output appears random, it shows that files may contain hidden or non-visible information.

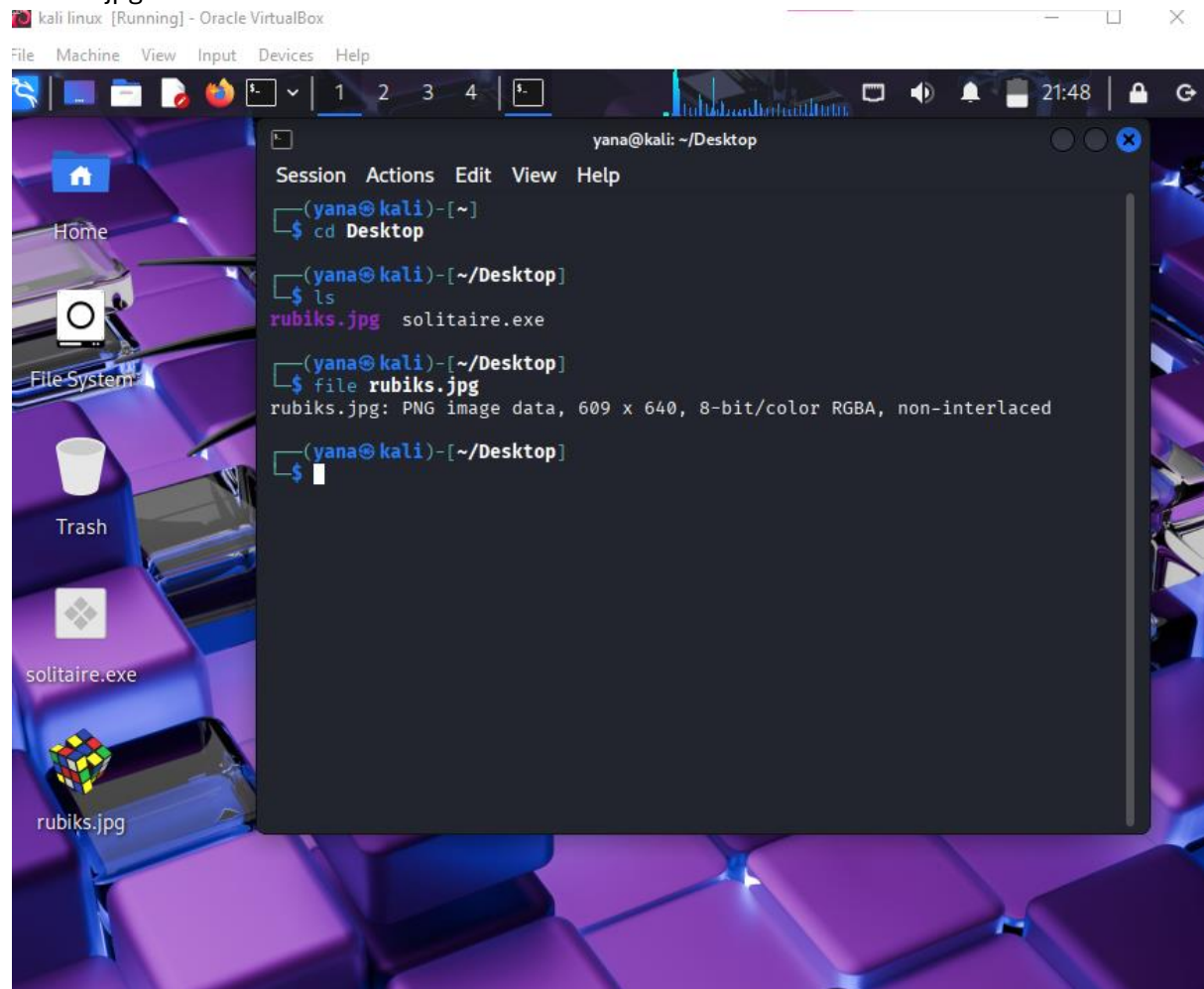
Solitaire.exe



The file command was used to analyze *solitaire.exe*. The result shows that the file is actually a PNG image, not a real executable file. This indicates that the file has been disguised by changing its extension, which can be misleading and potentially dangerous.

This demonstrates how attackers can hide the true nature of a file by renaming its extension.

Rubiks.jpg



The file command was used to analyze *rubiks.jpg*. The result shows that the file is actually a PNG image, not a JPEG as indicated by its extension. This suggests that the file has been renamed, which can be misleading and may hide its true format.